

Fairness in Machine Learning

Introduction

- The course of our lives is never fully one of our own making.
- Others' decisions affect our lives
 - Whether we get admission in a particular college
 - Whether a job is offered to us or not
 - Whether a mortgage is granted to us or not

Decision Making

- Machine Learning promises better decision making.
- By uncovering factors that humans may overlook
- By using the data themselves to find the factors (features) that bear a statistical relationship to the outcome.
- For example, object detection through examples.

Generalizing using ML

- Learning from examples requires that we provide the computer with good examples.
- Sufficiently diverse and well annotated examples etc.
- This means, evidence based system is only reliable as the evidence on which it is based.
- Being evidence based by no means ensures that it will lead to accurate, reliable, or fair decisions.

Generalizing using ML

- Especially true when ML is used to model human behavior and characteristics.
- Historical examples will almost always reflect historical prejudices.
- Finding patterns in these data will often replicate these very prejudices.

Predictive accuracy

- Human decision makers don't try to maximize predictive accuracy at all costs.
- For example, younger defendants are statistically more likely to re-offend, judges are loath to take this into account.
- Humans are also unlikely to make absurd decisions. Automated decision making would do it due to erroneous data.

ML in decision making

- In the criminal justice system, defendants are assigned scores to predict the risk of committing future crimes.
- Companies use ML to filter resumes of job applicants
- Statistical methods are also very common in lending, credit and insurance underwriting.

Demographic disparities

- Amazon uses data-driven system to determine the neighborhoods in which to offer free same-day delivery.
- A 2016 investigation found stark disparities in the demographic makeup of these neighborhoods.
- White residents were more than twice as likely as black residents to live in one of the qualifying neighborhoods.

Demographic disparities

- Disparities don't imply that the designer of the system intended for such inequalities to arise.
- Also, it is important to understand when disparities can be considered discrimination.
- Amazon argues that the system was designed for efficiency and cost considerations.
- The concern is it could contribute to the long-lasting cycles of inequality.

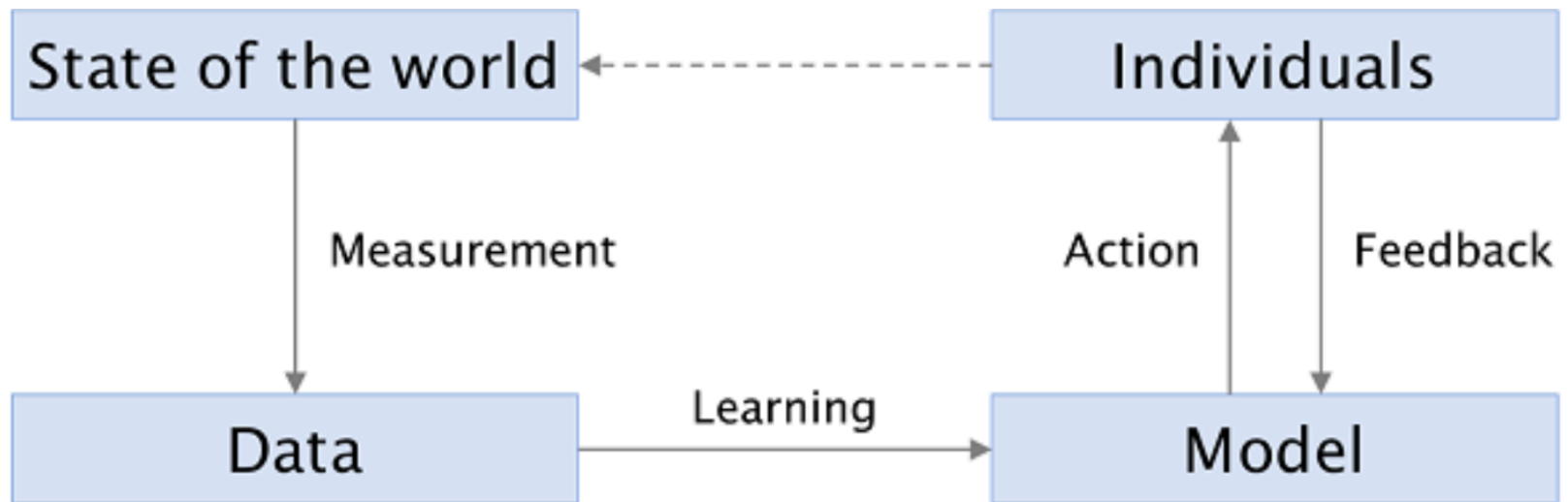
Demographic disparities

- The term *bias* is used to refer to demographic disparities in algorithmic systems that are objectionable for societal reasons.
- Statistical estimator is said to be biased when its expected value differs from the true value.
- This affects how we visualize ML systems and our responsibilities.

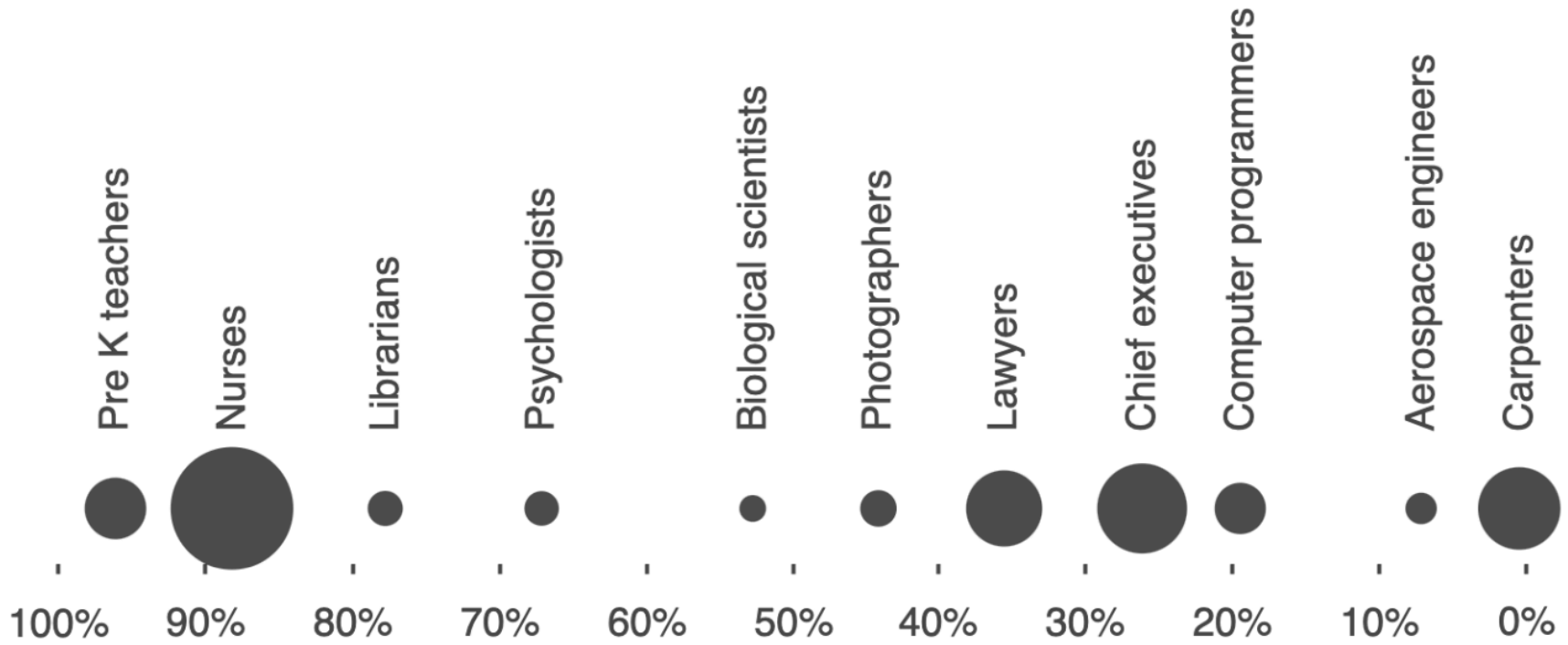
ML systems - design goals

- Is our goal to faithfully reflect the data?
- Do we have an obligation to question the data?, and
- Design the system to conform to some notion of equitable behavior.

ML loop



ML - Baseline data



ML systems - disparities

- Contributing factors - history of explicit discrimination, implicit attitudes, and stereotypes.
- We should make sure that ML doesn't become a part of this feedback loop.

Kaggle Competitions

- Top 30 competitions sorted by prize amount.
- 14 of these were tasks to make decisions about individuals.
- In most of these cases, there exist societal stereotypes or disparities that may be perpetuated by ML.

Kaggle Competitions

- Automated essay scoring task seeks to match human scores
- Students' linguistic choices are signifiers of social group membership.
- Humans can have prejudices based on such factors.
- Since human graders provide original labels....

Kaggle Competitions

- One of the competitions by Zillow, calls to improve Zestimate algorithm for predicting home sale prices.
- Any system predicting future price is likely to create a self-fulfilling feedback loop.
- Homes predicted to have lower sales price deter future buyers, suppressing demand and lowering final sale price.

Disparities in training

- Human society is full of demographic disparities and
- Training data will reflect these.

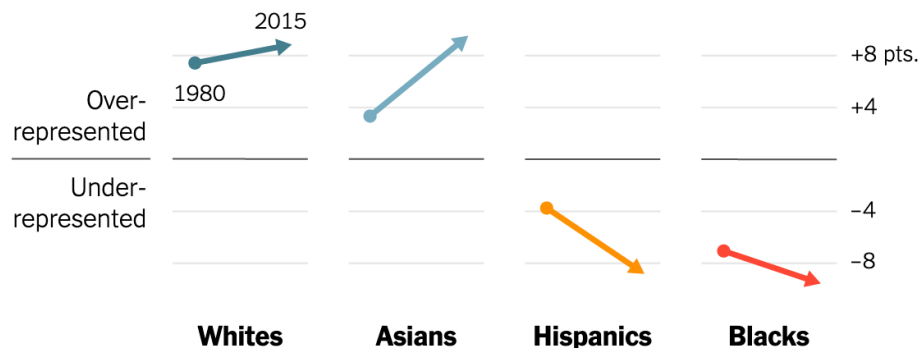
Measurement

- Measurement is fraught with subjective decisions and technical difficulties.
- For example, measuring demographic diversity of college campuses.

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Even With Affirmative Action, Blacks and Hispanics Are More Underrepresented at Top Colleges Than 35 Years Ago

By JEREMY ASHKENAS, HAEYOUN PARK and ADAM PEARCE AUG. 24, 2017



Measurement

- Authors argue gap has grown between college-age population and enrolled black and Hispanic freshmen.
- 100 American colleges from 1980 - 2015.
- Interestingly, multiracial category was introduced in 2008 and the article ignores comparisons.

Measurement

- How many students who might have checked the white or black box checked multiracial instead?
- How might this have affected the percentages of white and black students. Also,
- Would a person with black and latino parents identify as black in 2015 than in 1980s?

Measurement - summary

- Even a seemingly straightforward question about demographic diversity is impossible to answer,
- Without making assumptions.
- Also illustrates the difficulty of measurement.
- Measuring almost any human attribute is subjective and challenging.

Measurement - summary

- It becomes more chaotic when ML practitioners have to create categories.
- One of the areas where that is strictly true is defining target variable
- For example, will the defendant recidivate if released on bail?
- Will the candidate be a good employee?

Target

- Biases in the definition of the target variable are very critical.
- They are guaranteed to bias the predictions relative to the actual construct, we want to predict.
- Target variable is the hardest from a measurement standpoint.
- For example, credit worthiness was created to successfully extend credit.
- It is not an intrinsic property of people.

Target

- Example, let us say target variable is good employee.
- We may use performance review scores to quantify it.
- This means our data inherits any biases in managers' evaluations.

Target

- In some cases we may be more objective, at least in principle.
- In criminal risk assessment, training data could be who actually went on to commit a crime,
- Rather than judges' decision about bail.

Target

- We can't measure who committed a crime.
- We use a proxy, arrests.
- That means training data may contain distortions not due to prejudices of judges but may be due to discriminatory policing.

Target

- On the other hand, if our target is whether the defendant appears or fails to appear in court for trial,
- We would be able to measure it with perfect accuracy.
- In hiring, instead of relying on performance review (for example in a sales job) we might rely on number of sales.

Camera bias

- Visual works has infinite bandwidth, cameras' are limited.
- What is relevant and what is not can be a source of bias.
- Both film and digital cameras have been more adept at photographing lighter-skinned individuals.
- One reason being, default settings such as color balance which were optimized for lighter skin tones.

Camera bias

- Another reason is the limited dynamic range of cameras which makes it hard to capture brighter and darker tones in the same image.
- This started changing in 1970s due to complaints from furniture companies and chocolate companies.

Camera bias

- They complained about the difficulty of capturing details of furniture and chocolate.
- Another impetus came from increasing diversity of television subjects.

Images to datasets

- When we go from images to datasets of images, another layer of potential bias is introduced.
- For example, image sets for training object recognition systems.
- If these datasets represented the visual world, we may expect a CV system trained on one such dataset do well on another.

Images to datasets

- There is a drop in accuracy.
- This shows that datasets are biased relative to each other in a statistical sense.
- However, ML can mitigate some of these measurement biases.

ML mitigating biases (measurement)

- ML is making it possible to improve representation of tones in images.
- Diagnoses and treatments are sometimes personalized by race.
- It is used as a crude proxy for ancestry and genetics, and sometimes environmental and behavioral factors.

ML mitigating biases (measurement)

- If we can measure factors that are medically relevant and incorporate them, instead of race, into statistical models
- We may increase accuracy of diagnoses and treatments.

Data to Models

- Predictive models are good at calibration.
- Ensuring that predictions subsume all features in the data.
- Calibration also means we should expect our models to faithfully reflect disparities found in the input data.

Data to Models

- We may like to mine some patterns in the data, for example, smoking is associated with cancer,
- While other patterns (girls like pink and boys like blue), we may want to avoid learning.
- Learning algorithms have no way to distinguish between these two patterns.

Data to Models

- These are result of social norms and moral judgements.
- With lack of specific intervention ML will extract stereotypes, including incorrect and harmful ones.

Machine translation

The image shows two instances of the Google Translate interface, demonstrating how machine translation can misinterpret context. In the first instance, the English input is "She is a doctor. He is a nurse." and the Turkish output is "O bir doktor. O bir hemşire." (He is a doctor. He is a nurse). In the second instance, the English input is "O bir doktor. O bir hemşire" and the Turkish output is "He is a doctor. She is a nurse ✓".

Example 1:

English: She is a doctor.
He is a nurse.

Turkish: O bir doktor.
O bir hemşire.

Example 2:

Turkish: O bir doktor.
O bir hemşire

English: He is a doctor.
She is a nurse ✓

Data to Models

- When we build a statistical model of language from text
- We should expect gender associations of occupation words to match real world labor statistics.
- Example: Is it sufficient to withhold gender form data, if we are scoring resumes for a programming job?

Data to Models

- It is not that simple.
- Because of proxies and redundant encodings.
- There are other attributes that might correlate with gender.
- For example, in our society the age at which someone starts programming is correlated with gender.

Data to Models

- How long someone has been programming is valuable for the suitability of the programming job.
- Another reason why ML may perform worse for some groups is the sample size disparity.
- Our model test may report overall accuracy.
- This hides the fact about how the model performs for each group.

Anomaly detection

- This is one area where we find high error rates for minority groups.
- This is the idea of detecting behavior that deviates from the norm.
- Example, Nymwars controversy
- Where Google, Meta and other tech companies aimed to block users who used uncommon names (presumably fake).

Anomaly detection

- In some cultures most people receive names from a small set of names.
- In other cultures names may be more diverse and more likely unique.
- For users in the latter culture a popular name would be more likely to be fake.
- If we are not careful, learning algorithms will generalize based on majority culture.

Anomaly detection

- It could lead to higher error rate for minority groups.
- Attempting to avoid this by making the model complex can cause overfitting.
- One way to avoid this is to explicitly model differences between groups.
- There are technical and ethical challenges associated with this as well.

Model calibration

- A calibrated model faithfully captures patterns in the underlying data.
- Predictions using that model will have disparate error rates for different groups.
- If these groups have different base rates, i.e., rates of positive and negative outcomes.

Model calibration

- Understanding predictions requires understanding not just the model, but also the population differences between the groups.
- Population characteristics can shift over time.
- This machine learning phenomena is called drift.
- If subpopulations change differently over time, but the model isn't retrained, that can introduce disparities.

Correlation is not causation

- A major limitation in ML is that it reveals correlations.
- We often use its predictions as if they reveal causation.
- For example, an early ML system in healthcare famously learned that patients with asthma had lower risk of pneumonia.

Corrleation is not causation

- This was a true pattern in the data, but the likely reason was that asthmatic patients were more likely to receive in-patient care.
- So it is not valid to use the prediction to decide whether or not to admit a patient.
- Same principle is seen in the use of ML for predicting traffic congestion.

Correlation is not causation

- If sufficiently many people choose their routes based on the prediction, then
- The route predicted to be clear will in fact be congested.
- The prediction could also reinforce the outcome resulting in feedback loops.

Feedback and loops

- Many systems receive feedback.
- When search engine serves results, it records the links that the users clicks.
- And how long the user spends on those pages.
- It treats that as implicit signals about which results to be most relevant.

Feedback and loops

- When a video sharing website recommends a video is used thumbs up / down feedback as an explicit signal.
- But feedback is tricky to interpret correctly.
- If a user clicked on the first link of search results, is that simply because it was first, or
- Because it was in fact the most relevant.
- This is a case of action affecting the outcome.

Feedback and loops

- Bias in feedback might also reflect cultural prejudices, which is harder to characterize.
- For example, clicks on targeted ads that appear alongside search results might reflect gender and racial stereotypes.
- There is a study that hints
- That Google searches for Black sounding names such as Latanya Farrell were much more likely to result in ads for arrest records.

Feedback and loops

- Than searches for white sounding names. (Kristen Haring)
- One potential explanation is, users are more likely to click on ads that conform to stereotypes.
- And advertising is optimized for maximizing clicks.
- Even feedback can lead to unexpected or undesirable biases.

Feedback and loops

- Another example:
- Predictive policing determines certain areas of a city to be at high risk for crime.
- More police officers may be deployed.
- Officers in areas might be subtly lowering their threshold for stopping, searching or arresting people, perhaps even unconsciously.

Feedback and loops

- Either way the prediction will appear to be validated even if it had been made purely on data biases.